GLOBAL SOLAR ATLAS
BY WORLD BANK GROUP

Quezon City

14.733114°, 121.069864°
Fe Street, Quezon City, Philippines
Time zone: UTC+08, Asia/Manila [PST]

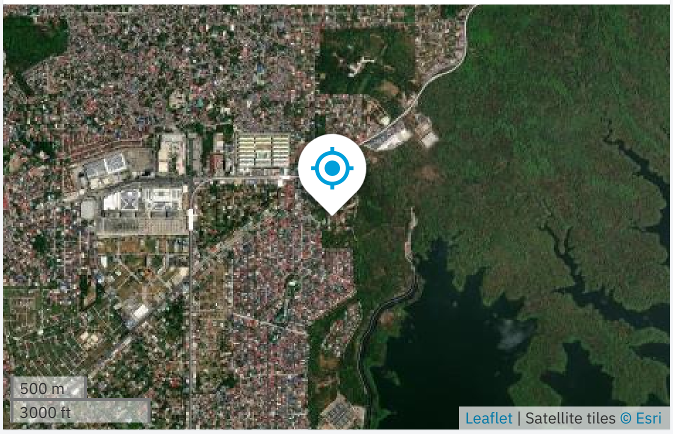
Report generated: 16 Jun 2026

SITE INFO

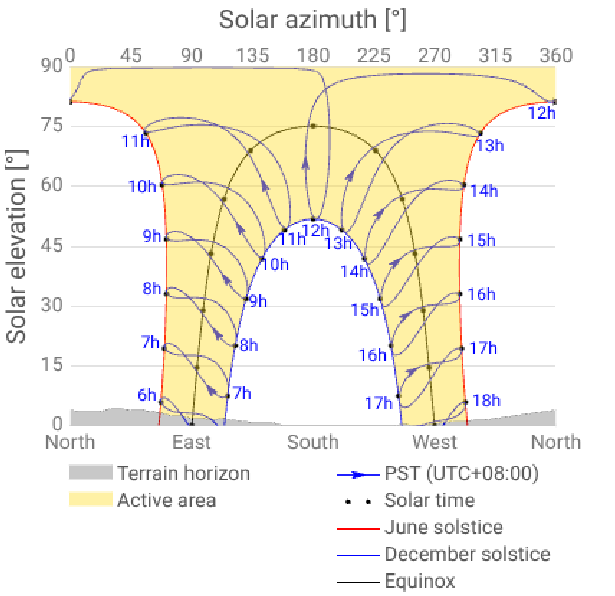
Map data

Map data			Per year
Direct normal irradiation	DNI	**1247.2**	kWh/m²
Global horizontal irradiation	GHI	**1704.5**	kWh/m²
Diffuse horizontal irradiation	DIF	**848.3**	kWh/m²
Global tilted irradiation at optimum angle	GTI opta	**1747.7**	kWh/m²
Optimum tilt of PV modules	OPTA	**14 / 180**	°
Air temperature	TEMP	**26.3**	°C
Terrain elevation	ELE	**99**	m

Map



Horizon and sunpath



PVOUT map

© 2026 The World Bank Group

PV ELECTRICITY AND SOLAR RADIATION

PV system configuration

Pv system: **Medium size comercial**

Azimuth of PV panels: **356º**

Tilt of PV panels: **5º**

Installed capacity: **123.5 kWp**

Annual averages

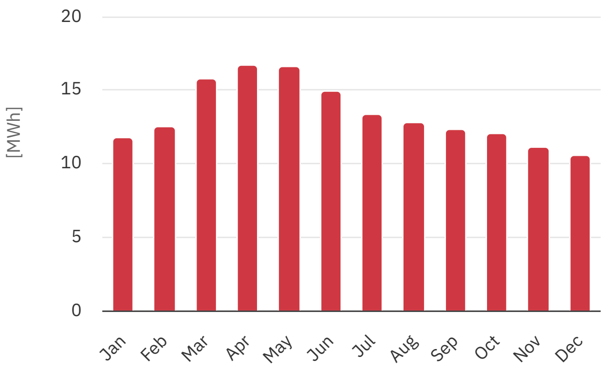
Total photovoltaic power output and Global tilted irradiation

160.436 MWh per year

1675.3 kWh/m² per year

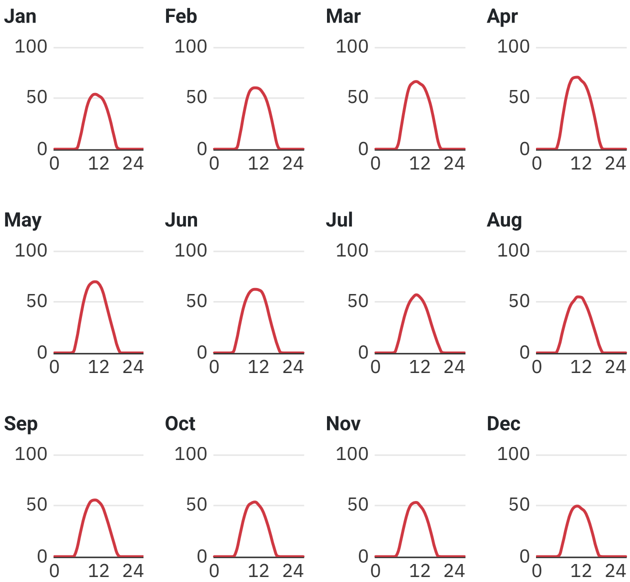
Monthly averages

Total photovoltaic power output



Average hourly profiles

Total photovoltaic power output [kWh]



Average hourly profiles

Total photovoltaic power output [kWh]

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
0 - 1												
1 - 2												
2 - 3												
3 - 4												
4 - 5												
5 - 6				0.025	0.717	0.691	0.308					
6 - 7	0.755	1.090	3.922	9.781	13.673	12.515	9.309	7.169	6.994	6.234	4.138	1.325
7 - 8	12.211	15.641	23.285	32.337	33.967	30.595	24.428	22.148	23.355	22.533	18.984	13.631
8 - 9	29.434	35.405	44.210	52.536	52.013	46.075	38.422	35.400	38.520	38.213	35.028	29.286
9 - 10	44.306	51.450	59.709	65.398	63.665	55.995	48.374	45.829	48.950	48.812	47.045	41.786
10 - 11	51.738	58.717	64.771	69.978	68.397	61.016	53.961	51.132	54.518	52.310	52.079	48.123
11 - 12	53.749	60.049	66.150	70.416	69.605	62.210	56.822	54.826	55.393	53.415	52.987	49.423
12 - 13	52.122	59.332	64.086	67.138	67.559	61.581	54.406	54.379	53.442	50.448	49.742	47.041
13 - 14	49.001	55.608	61.005	63.169	60.451	58.492	49.283	48.618	48.534	45.154	44.618	43.565
14 - 15	41.189	48.714	52.964	54.376	47.168	48.167	40.535	40.059	38.393	35.757	35.509	35.062
15 - 16	29.156	36.495	40.695	40.733	33.068	33.546	28.614	28.916	26.413	23.658	22.563	22.790
16 - 17	13.970	20.028	23.025	23.827	19.563	19.945	17.374	17.214	13.895	10.227	8.272	8.907
17 - 18	1.280	4.031	5.710	6.248	6.250	7.787	7.329	5.659	2.576	0.662	0.252	0.428
18 - 19						0.283	0.371					
19 - 20												
20 - 21												
21 - 22												
22 - 23												
23 - 24												
Sum	379	447	510	556	536	499	430	411	411	387	371	341

THE WORLD BANK IBRD · IDA
ESMAP Energy Sector Management Assistance Program
SOLARGIS
© 2026 The World Bank Group

GLOBAL SOLAR ATLAS
BY WORLD BANK GROUP

PV ELECTRICITY AND SOLAR RADIATION

Annual averages

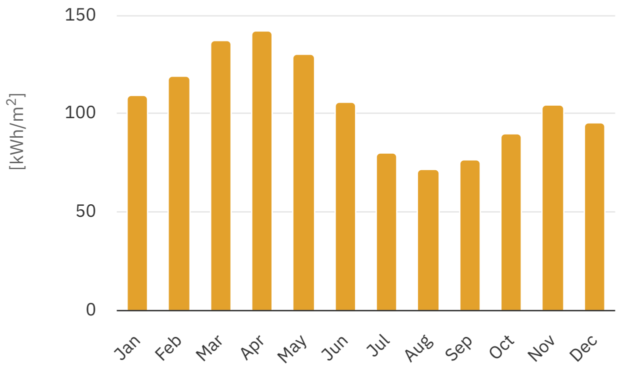
Direct normal irradiation

1260.8

kWh/m² per year

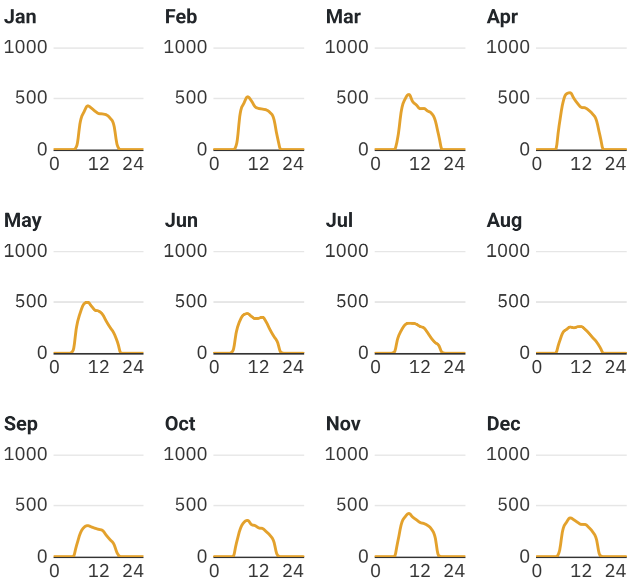
Monthly averages

Direct normal irradiation



Average hourly profiles

Direct normal irradiation [Wh/m²]



UTC+08

Average hourly profiles

Direct normal irradiation [Wh/m²]

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
0 - 1												
1 - 2												
2 - 3												
3 - 4												
4 - 5												
5 - 6					24	22	8					
6 - 7	36	54	123	235	262	216	146	102	114	138	150	44
7 - 8	275	357	390	454	399	324	226	199	230	271	325	261
8 - 9	370	454	497	545	482	377	278	232	287	336	391	337
9 - 10	427	516	539	555	498	385	293	255	303	354	422	379
10 - 11	405	477	470	500	459	360	291	247	290	315	387	360
11 - 12	374	419	436	450	419	338	283	257	277	304	361	335
12 - 13	352	402	401	413	410	342	260	258	266	281	335	316
13 - 14	348	396	402	408	377	351	247	233	255	275	324	315
14 - 15	340	388	378	384	311	304	202	196	209	245	306	286
15 - 16	308	363	357	349	252	231	147	154	166	210	273	245
16 - 17	243	306	294	292	198	168	105	114	124	153	198	178
17 - 18	38	126	142	140	109	110	77	58	31	18	13	15
18 - 19						6	6					
19 - 20												
20 - 21												
21 - 22												
22 - 23												
23 - 24												
Sum	3,515	4,258	4,429	4,726	4,200	3,535	2,570	2,306	2,554	2,898	3,484	3,070

© 2026 The World Bank Group

GLOSSARY

Acronym	Full name	Unit	Type of use
DIF	Diffuse horizontal irradiation	kWh/m², MJ/m²	Average yearly, monthly or daily sum of diffuse horizontal irradiation (© 2025 Solargis)
DNI	Direct normal irradiation	kWh/m², MJ/m²	Average yearly, monthly or daily sum of direct normal irradiation (© 2025 Solargis)
ELE	Terrain elevation	m, ft	Elevation of terrain surface above/below sea level, processed and integrated from SRTM-3 data and related data products (SRTM v4.1 © 2004 - 2025, CGIAR-CSI)
GHI	Global horizontal irradiation	kWh/m², MJ/m²	Average annual, monthly or daily sum of global horizontal irradiation (© 2025 Solargis)
GTI	Global tilted irradiation	kWh/m², MJ/m²	Average annual, monthly or daily sum of global tilted irradiation (© 2025 Solargis)
GTI_opta	Global tilted irradiation at optimum angle	kWh/m², MJ/m²	Average annual, monthly or daily sum of global tilted irradiation for PV modules fix-mounted at optimum angle (© 2025 Solargis)
OPTA	Optimum tilt of PV modules	°	Optimum tilt of fix-mounted PV modules facing towards Equator set for maximizing GTI input (© 2025 Solargis)
PVOUT_total	Total photovoltaic power output	kWh, MWh, GWh	Yearly and monthly average values of photovoltaic electricity (AC) delivered by the total installed capacity of a PV system (© 2025 Solargis)
PVOUT_specific	Specific photovoltaic power output	kWh/kWp	Yearly and monthly average values of photovoltaic electricity (AC) delivered by a PV system and normalized to 1 kWp of installed capacity (© 2025 Solargis)
TEMP	Air temperature	°C, °F	Average yearly, monthly and daily air temperature at 2 m above ground. Calculated from outputs of ERA5 model (© 2025 ECMWF, post-processed by Solargis)

ABOUT

This pdf report (the “Work”) is automatically generated from the Global Solar Atlas online app (https://globalsolaratlas.info/), prepared by Solargis under contract to The World Bank, based on a solar resource database that Solargis owns and maintains. It provides the estimated solar resource, air temperature data and potential solar power output for the selected location and input parameters of a photovoltaic (PV) power system.

Copyright © 2026 The World Bank
1818 H Street NW, Washington DC 20433, USA

The World Bank, comprising the International Bank for Reconstruction and Development (IBRD) and the International Development Association (IDA), is the commissioning agent and copyright holder for this Work, acting on behalf of The World Bank Group. The Work is licensed by The World Bank under a Creative Commons Attribution license (CC BY 4.0 IGO) with a mandatory and binding addition (please refer to the GSA website for full terms and conditions of use https://globalsolaratlas.info/support/terms-of-use).

The World Bank Group disclaims all warranties of any kind related to the provision of the Work.

The Work is made available solely for general information purposes. Neither the World Bank, Solargis nor any of its partners and affiliates hold the responsibility for the accuracy and/or completeness of the data and shall not be liable for any errors, or omissions. It is strongly advised that the Work be limited to use in informing policy discussions on the subject, and/or in creating services that better educate relevant persons on the viability of solar development in areas of interest. As such, neither the World Bank nor any of its partners on the Global Solar Atlas project will be liable for any damages relating to the use of the Work for financial commitments or any similar use cases. Solargis has done its utmost to make an assessment of solar climate conditions based on the best available data, software, and knowledge.

Sources: Solar database and PV software © 2026 Solargis